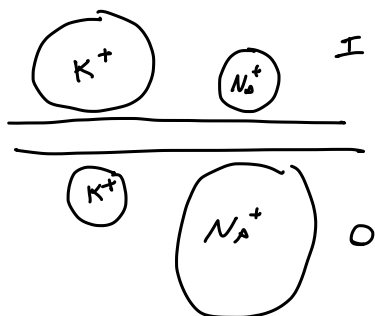


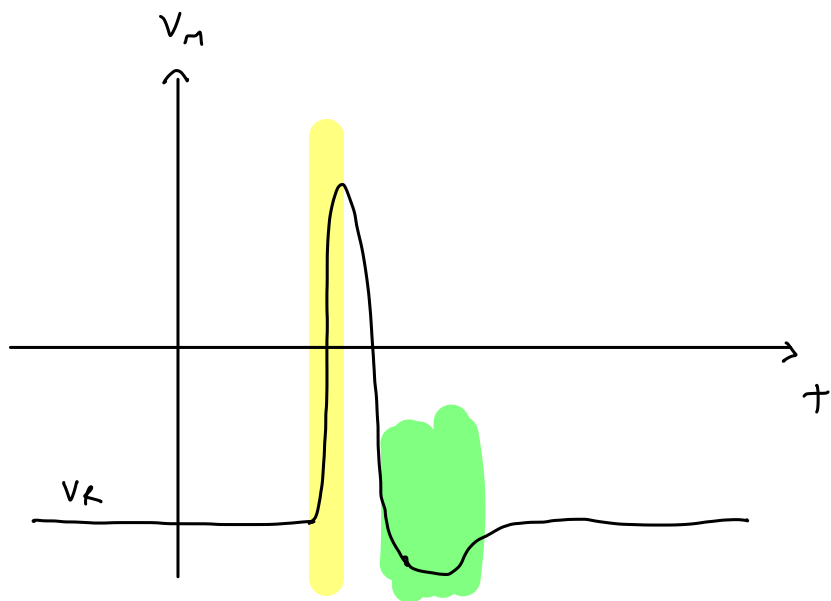
11/5

$$G_{Na}(V_m, t)$$

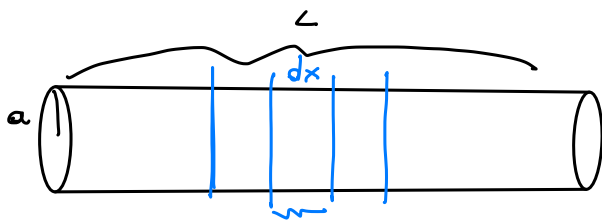
$$G_K(V_m, t)$$



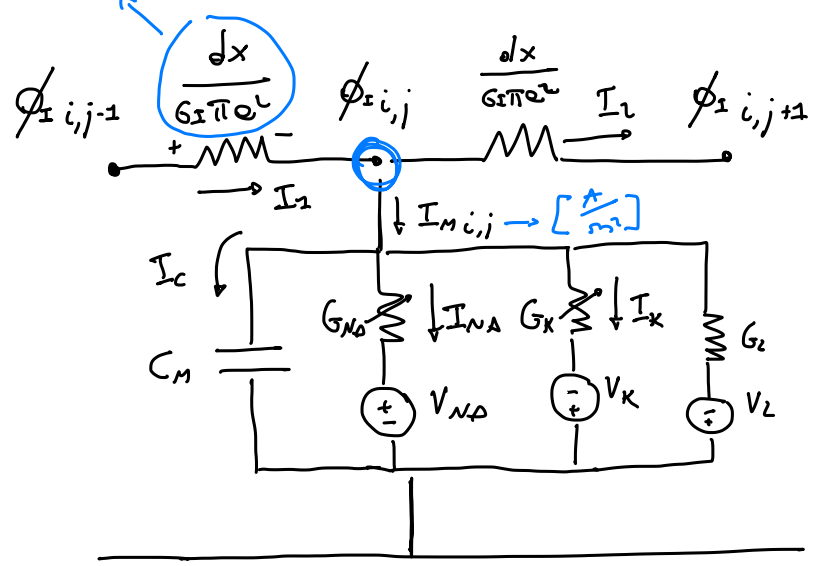
$$\left. \begin{array}{cc} [Na]_I & [Na]_O \\ [K]_I & [K]_O \end{array} \right\} \text{constant}$$



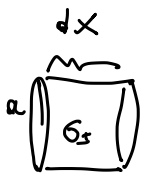
$\angle \gg \circ$



$[Z] \rightarrow I_1 [A]$ $\rightarrow$ circuit... $i \rightarrow \text{time} \quad j \rightarrow \text{space}$



ϕ_0



$$\frac{1}{G_1} \frac{dx}{\pi Q^2}$$



$$R_0 \ll \frac{dx}{61 \pi Q^2}$$

$$\Rightarrow \phi_0 = C_{\text{out}} \pi$$

$$I_1 = \frac{\phi_{i,j-1} - \phi_{i,j}}{dx} G_{i,j} \pi Q^2$$

$$I_2 = \frac{\phi_{i,j} - \phi_{i,j+1}}{dx} G_{i,j} \pi Q^2$$

$$I_1 = I_2 + I_{M,i,j} 2\pi Q dx$$

$$\left[\frac{\phi_{i,j-1} - \phi_{i,j}}{dx} - \frac{\phi_{i,j} - \phi_{i,j+1}}{dx} \right] G_{i,j} \pi Q^2 = I_{M,i,j} 2\pi Q dx$$

$$I_{M,i,j} \left(\frac{2}{G_{i,j}} \right) = \frac{\phi_{i,j-1} + \phi_{i,j+1} - 2\phi_{i,j}}{dx^2}$$

$$dx \rightarrow \phi$$

$$\frac{\partial^2 \phi(x,t)}{\partial x^2} = I_M(x,t) \frac{2}{G_{i,j}}$$

GENERAL!

$$I_M = I_C + I_{ionic} = C_M \frac{dV_M}{dt} + I_{ionic}$$

$$\frac{\partial^2 \phi_I}{\partial x^2} = \frac{2}{G_I Q} \left(C_M \frac{\partial V_M}{\partial t} + I_{ionil} \right)$$

$$\frac{\partial V_M}{\partial t} = \frac{1}{C_M} \left(\frac{G_I Q}{2} \frac{\partial^2 \phi_I}{\partial x^2} - I_{ionil} \right) \quad \text{GENERAL}$$

hyp $\rightarrow \phi_0 = \cos T$

$$V_M = \phi_I - \phi_0$$

$$\frac{\partial^2 V_M}{\partial x^2} = \frac{\partial^2 \phi_I}{\partial x^2}$$

H.H

$$\frac{\partial V_M(x,t)}{\partial t} = \frac{1}{C_M} \left(\frac{G_I Q}{2} \frac{\partial^2 V_M(x,t)}{\partial x^2} - I_{ion}(x,t) \right)$$

$\rightarrow OK \rightarrow \phi_0 A$

$\rightarrow NO \rightarrow$ STIMULATION

NERVE CUFF STIM. $\rightarrow$ FES
WITH EXT. CURRENT

INITIAL
CONDITIONS
GATING
VARIABLES

$$\pi_0 = \frac{\alpha_N(V_R)}{\alpha_N(V_R) + \beta_N(V_R)}$$